

Programming 2025/2026

Laboratory

Session 1 (April 14)

Strings

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Sequence of Values

- **Immutable**: contents cannot be modified once created
 - **string**: a sequence of characters, e.g. `"Name"`
 - **tuple**: a sequence of elements placed in parentheses, separated by comma, e.g. `('Mario', 'Rossi', 'M', 20)`
 - **range**: a sequence of integers in a range [a,b), e.g. `range(1,5)` for `1..4`
- **Mutable**: contents can be modified
 - **lists**
 - **dictionaries**
 - ...

String

- A string can be defined with a single quote marks ' ' or a double quote marks " "
- A string is an immutable sequence

```
s = "time"  
s[0] = 'F'
```

```
Traceback (most recent call last):  
  File "<stdin>", line 1, in <module>  
TypeError: 'str' object does not support item assignment
```

- String operations
 - We already know concatenate +, repetition *, **len()**, selection []
 - We have more
 - >, <, ==, slicing operators [:], [::], **in** and **not in** operators

String Slicing

- We already know how to access a single character in a string (e.g. `s[0]`)
- The slicing operator (`[ : ]`) extracts a substring indicated by a range of indices
 - Is used as `[start_index:final_index]`
 - `start_index` and `final_index` are integers
 - `final_index` is not included, the selection finishes at `final_index-1`
 - If `start_index` is not specified (`[ :final_index]`), the selection starts from the beginning of the string
 - If `final_index` is not specified (`[start_index: ]`), the selection continues until the end of the string (included)

-6	-5	-4	-3	-2	-1							
0	1	2	3	4	5							
+---+---+---+---+---+---+												
	A		n		d		r		e		a	
+---+---+---+---+---+---+												

```
>>> name = "Andrea"
>>> name[1:2]
'n'
>>> name[:2]
'An'
>>> name[1:]
'ndrea'
```

String Slicing

- We already know how to access a single character in a string (e.g. `s[0]`)
- The slicing operator (`[ : ]`) extracts a substring indicated by a range of indices
 - Can also be used as `[start_index:final_index:step]`
- The element at every `step` from `start_index` to `final_index-1` is selected (default value is 1)
- If `step` is negative, the slicing is similar but is done in the reverse order

-6	-5	-4	-3	-2	-1							
0	1	2	3	4	5							
+---+---+---+---+---+---+												
	A		n		d		r		e		a	
+---+---+---+---+---+---+												

```
>>> name[0:4:2]
'Ad'
>>> name[-5:-1:2]
'nr'
>>> name[4:0:-2]
'ed'
>>> name[-2:-5:-2]
'ed'
```

Exercise (3.1)

- Write a program that prints a given string of characters in reverse order (e.g. if the string is "time", the program will print "emit")

Exercise (3.2)

- Write a program that given two strings, prints them in alphabetical order or prints only one of them if they have the same content

The `in` Operator

- How can I check if an element is contained in a sequence?
 - We can use the operators `in` or `not in`
- What are the results of the following expressions?

```
'e' in "Andrea"  
"nd" in "Andrea"  
'p' not in "Andrea"  
'E' in "Andrea"
```

- The `string` module contains some useful constants

```
string.ascii_letters  
string.ascii_lowercase  
string.ascii_uppercase  
string.digits  
string.punctuation  
string.whitespace
```